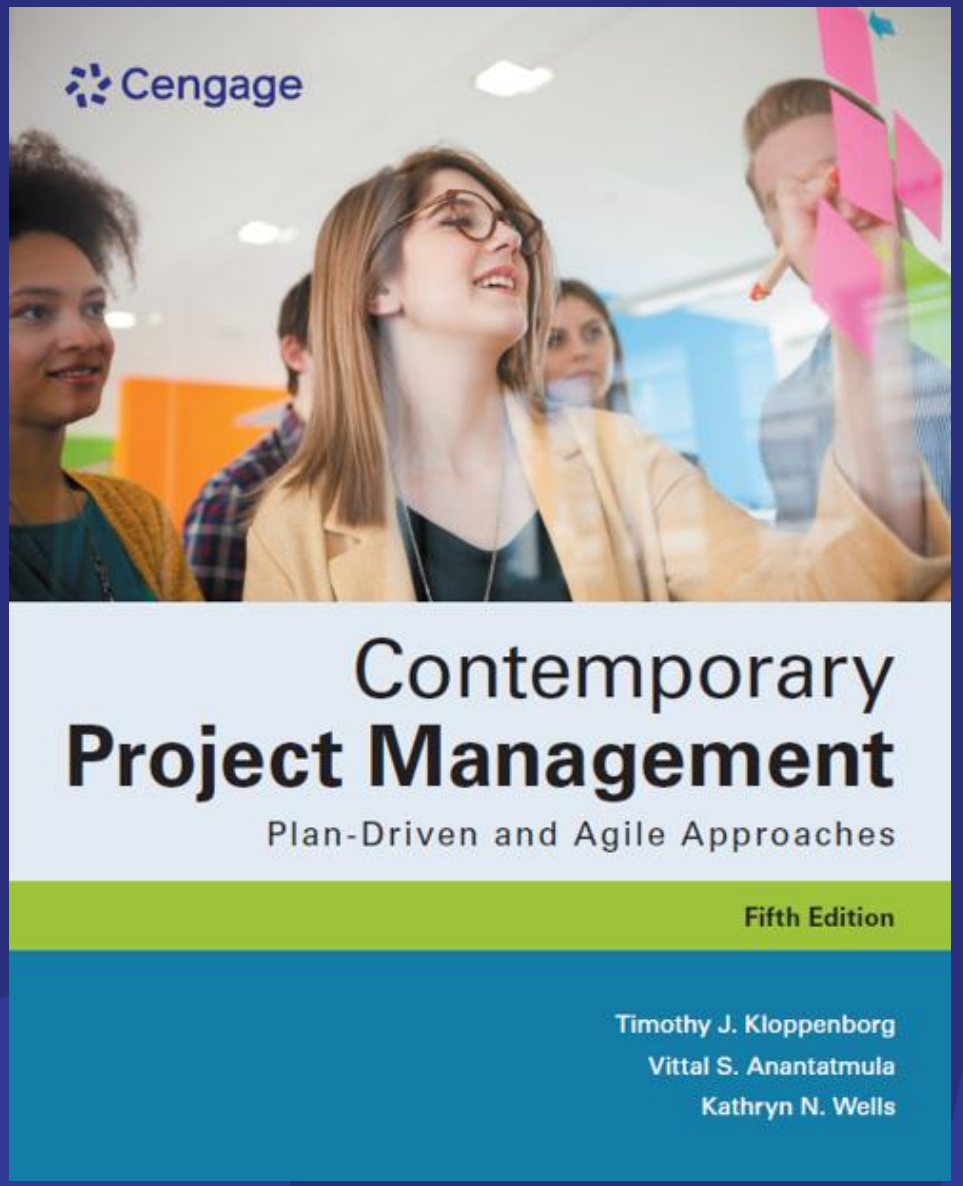


Chapter 14

Determining Project Progress and Results

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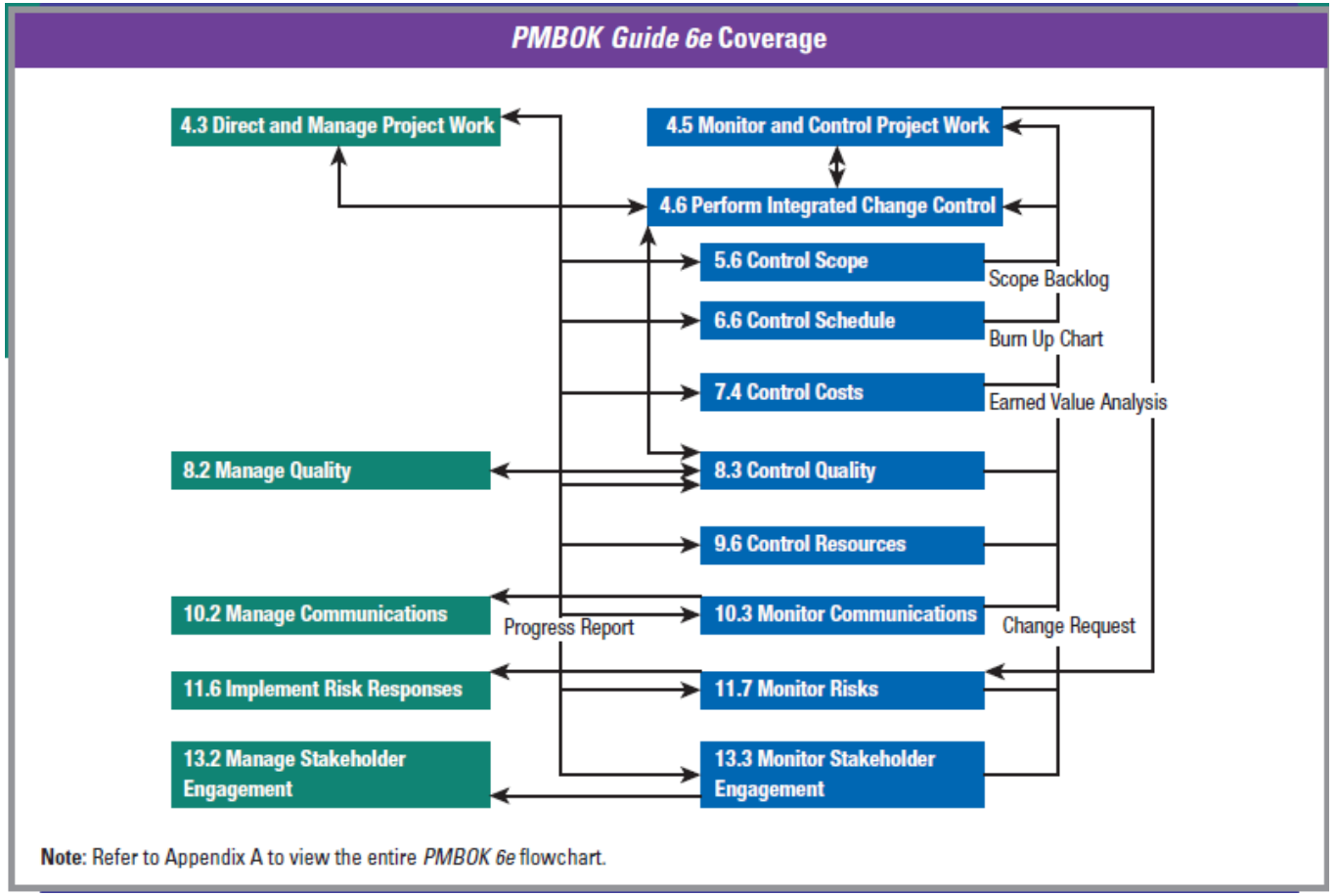
Icebreaker

What are two things you learned from Project Procurement and Partnering?



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Determining Project Progress and Results (1 of 2)



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Determining Project Progress and Results (2 of 2)

Overview of the Book
14.1 Project Balanced Scorecard Approach
14.2 Internal Project Issues
14.3 Customer Issues
14.4 Financial Issues
14.5 *PMBOK Guide 7e* Determining Project Progress and Results
14.6 Agile
14.7 Using MS Project to Monitor and Control Projects
14.8 Replanning If Necessary
Summary

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Core Objectives

- **14-1** Develop and demonstrate use of a change control system.
- **14-2** Demonstrate how to monitor and control project risks with various resolution strategies.
- **14-3** Create and present a project progress report.
- **14-4** Describe the importance of formal project progress reporting and communications.
- **14-5** Demonstrate negotiating skills.
- **14-6** Manage conflicts throughout the project execution.
- **14-7** Describe project quality control tools, including how and when to use each.
- **14-8** Calculate current project schedule and budget progress, and predict future progress, using Earned Value Analysis.
- **14-9** Document project progress using MS Project.

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Agile Objectives

- **14-10** Create an Agile iteration plan.
- **14-11** Conduct Agile stand-up meetings and document meeting minutes.
- **14-12** Demonstrate use of Kanban boards.
- **14-13** Capture and share lessons learned (retrospectives).

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14.1 Project Balanced Scorecard Approach

Five Aspects of Project Determination
The Project Manager and Decision Making
The Balanced Scorecard Approach

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Five Aspects of Project Determination

“Determine” can mean:

1. *to give direction to or decide the course of.*
2. *to be the cause of, to influence, or to regulate.*
3. *to limit in scope.*
4. *to reach a decision.*
5. *to come to a conclusion or resolution.*

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The Project Manager and Decision Making

During project execution, a project manager may need to:

- Give directions.
- Re-plan.
- Influence.
- Guard the scope.
- Follow-up to ensure proper results are delivered.

Project manager decision making may involve:

- Personally making decisions.
- Delegating decisions.
- Being part of a group that makes decisions.
- Facilitating a decision-making process.

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Project Balanced Scorecard Approach

To successfully accomplish all five aspects of project determination in managing project progress:

direct, regulate, limit, decide, and conclude

a project manager can think in terms of a balanced scorecard approach.

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Exhibit 14.1

Balanced Scorecard Approach to Project Determination

Internal Project	Customer	Financial
Direct and manage project work	Manage quality	Control resources
Monitor and control project work	Control quality	Control schedule
Perform integrated change control	Control scope	Control costs
Implement risk responses		
Monitor risks		
Manage communication		
Monitor communication		

Source: Adapted from Kevin Devine, Timothy J. Kloppenborg, and Priscilla O'Clock, "Project Measurement and Success: A Balanced Scorecard Approach," *Journal of Healthcare Finance* 36 (4) (2010): 38–50.

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14.2 Internal Project Issues

14.2a Direct and Manage Project Work
14.2b Monitor and Control Project Work
14.2c Monitoring Project Risk
14.2d Implement Risk Responses
14.2e Manage Communications
14.2f Monitor Communications

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14.2a Direct and Manage Project Work

Directing and managing project work is performing the work as defined in various components of the project management plan, including approved changes, with an intent to accomplish project objectives.

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Exhibit 14.2

Project Trade-Off Decisions at Tatro, Inc.

Tatro, Inc., is a company that describes itself as a designer, builder, and caretaker of fine landscaping. It has both commercial and private (homeowner) clients. Landscaping projects for private homes often cost well over $100,000. Homeowners who contract for landscaping projects of this magnitude are people who will not change their mind once they decide they want something special. These clients tend to focus closely on the process of a project. They wish to have polite, skilled workers with no interruptions. The reason they wish to have the project completed is to create a “wow factor.” Therefore, they will rarely compromise at all on either scope or quality, but they will often compromise on the necessary cost and schedule.

Source: Chris Tetrault, president, Tatro, Inc. Reprinted with permission.

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Exhibit 14.3

Useful Owner Representative Skills in Project Trade-Off Decision Making

Partnership	Building trust Improving relations Collaborating Creating alliances Assuring quality
Management	Planning Managing change Aligning resources
Leadership	Communicating Team building
Technical	Project management Knowledge of criteria

Source: Adapted from Denis R. Petersen, and E. Lile Murphree, Jr., "The Impact of Owner Representatives in a Design-Build Construction Environment," *Project Management Journal* 35 (3) (September 2004): 35–36.

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14.2b Monitor and Control Project Work

Monitoring and controlling project work includes activities such as identifying work packages for tracking, reviewing, and documenting project progress.

Monitoring

Reviewing the progress and capturing project performance data.

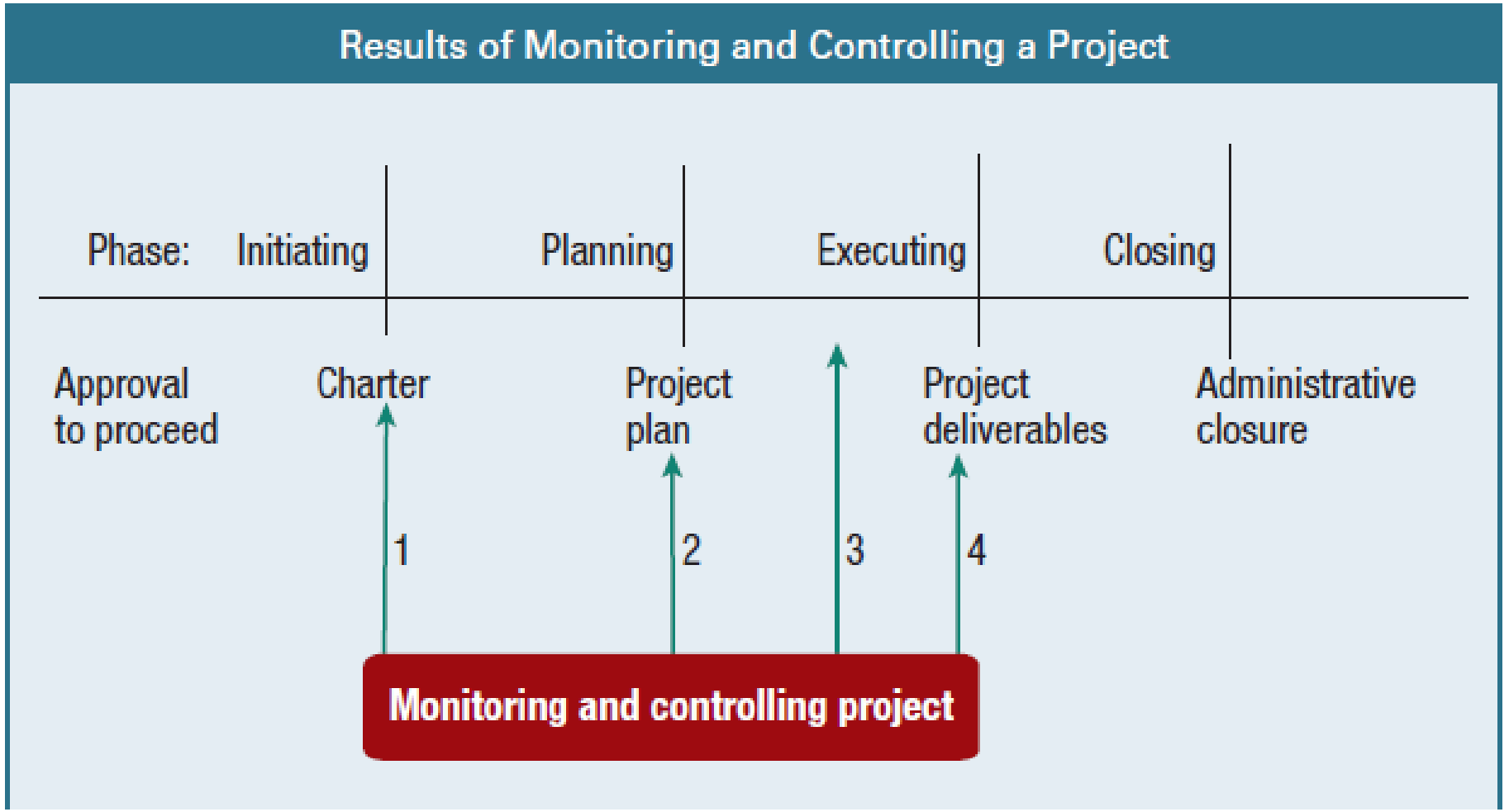
Controlling

Assessing actual performance obtained from monitoring.

*Note: A **variance** is a measurable departure from a planned baseline or expected value.*

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Exhibit 14.4



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Monitoring and Controlling Decisions

IF . . .	THEN . . .
actual progress is very different from the original intent	*the project charter needs to be revisited to ensure that the project still makes sense.*
progress is somewhat different from what was planned but the charter is still a good guide	*the project plan needs to be adjusted.*
the project plan is still a useful guide and progress has minor differences	*adjustments can be made in day-to-day instructions within the project executing stage.*
the results indicate the customer is ready to accept the project deliverables	*it is time to proceed into the project closing stage.*

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14.2b Perform Integrated Change Control

Perform Integrated Change Control is reviewing all change proposals, estimating their impact on project goals wherever appropriate, approving or declining changes, and managing changes to deliverables, schedules, budgets, and the project management plan.

Change Control

A process wherein change proposals to various project planning elements are acknowledged, formally documented, and approved or declined after review.

Change Control Board

A formal group authorized and responsible for reviewing, evaluating, approving, delaying, or rejecting any changes to any aspect of the project plan.

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14.2c Monitoring Project Risk

Monitor Project Risks is the process of adhering to the risk response plan of tracking identified risks, identifying new risks, monitoring residual risks, and evaluating the effectiveness of the risk response process throughout the project.

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Monitoring Project Risk Effort

Efforts needed to monitor risks include:

- tracking the identified risks.
- executing the response plans.
- evaluating their effectiveness.

Two categories of project management methods can help to deal with previously unidentified risks.

1) Recognize “unknown” risks *(they may add a contingency reserve of time, budget).*
2) Consider level of project team control *(to mitigate risk).*

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Exhibit 14.5

Risk Event Resolution Strategies

Risks Within Project Control	Risks Partially Within Project Control	Risks Outside Project Control
Understand and control WBS	Establish limits to customer expectations	Understand project context and environment
Closely monitor and control activity progress	Build relationships by understanding project from client's perspective	Actively monitor project environment
Closely manage all project changes	Use honesty in managing client expectations	Understand willingness or reluctance of stakeholders to agree to changes
Document all change requests	Work with client to reprioritize cost, schedule, scope, and/or quality	
Increase overtime to stay on schedule	Carefully escalate problems	
Isolate problems and reschedule other activities	Build team commitment and enthusiasm	
Research challenging issues early		

Source: Adapted from Hazel Taylor, "Risk Management and Problem Resolution Strategies for IT Projects: Prescription and Practice," *Project Management Journal* 37 (5) (December 2006): 55–60.

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14.2d Implement Risk Responses

Implement Risk Responses is the process where when a risk event occurs or is quite likely to occur soon, the person assigned to that risk executes the strategy identified in the risk management plan.

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14.2e Manage Communications

Manage Communications is all the work connected with the project communications plan, starting with planning for it; generating it; organizing and sharing it; and, storing and disposing of it.

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Collect Information on Executed Work and WIP

- How well is this particular activity proceeding in terms of time and budget?
- How well is the entire project proceeding in terms of time and budget?
- How much more money will need to be spent to complete the project?
- To what extent does the quality of this work meet requirements?
- How many hours of human resources were used to complete this activity, compared to the estimate?
- What methods are worth repeating for future project activities and projects?
- What methods require improvement before we do that type of work again?

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Report Performance

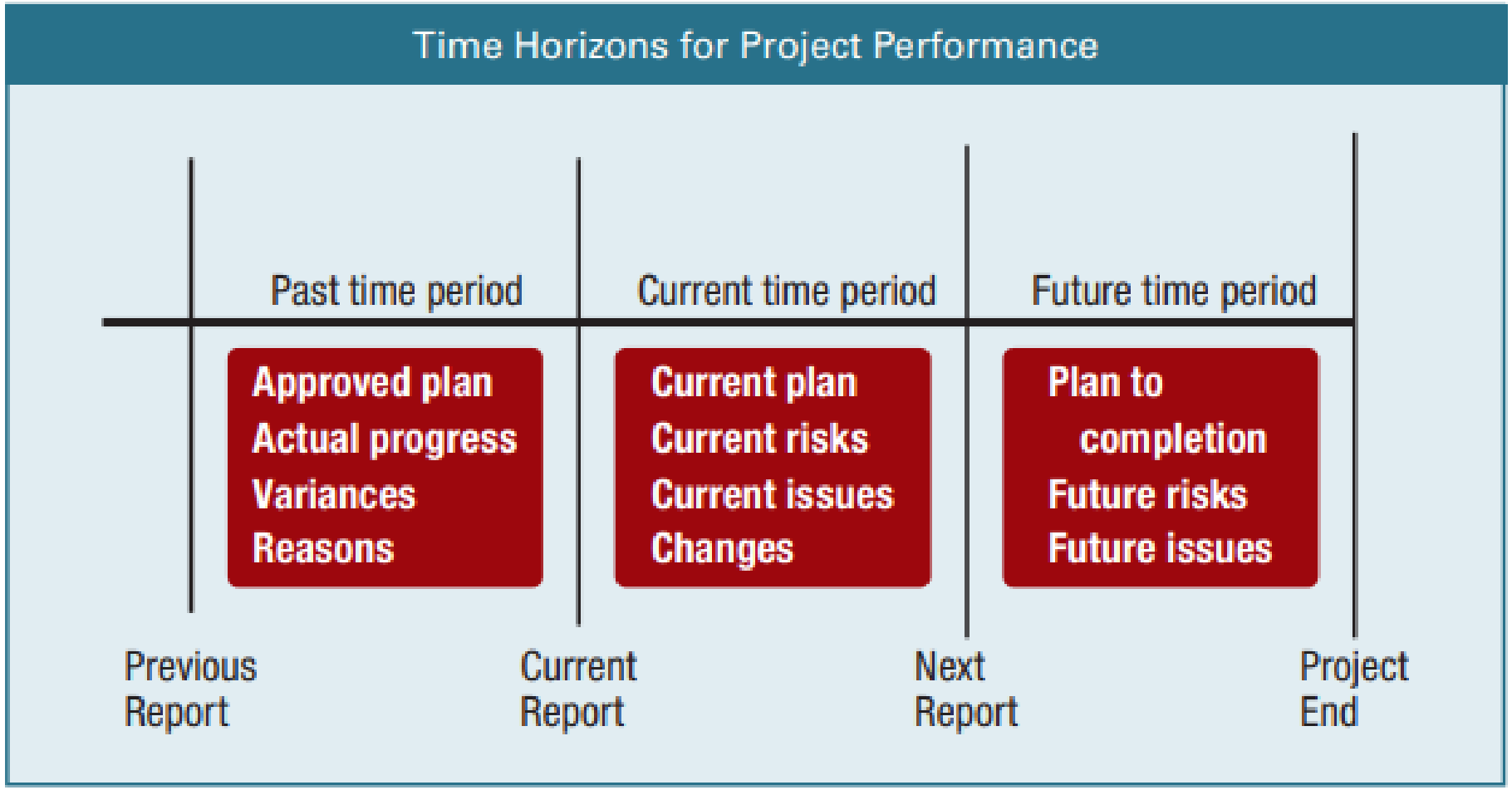
Work Performance Data
is used to create work performance information and reports.

Work Performance Information
is the information from performance data, collected from these processes, analyzed in context, and then integrated, considering relations across areas.

Work Performance Reports
are the compilation of work performance information in some physical or electronic form.

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Exhibit 14.6



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Progress Reporting Time Horizons

Past Time Period
is the immediate period between one's last report and now.

Current Time Period
is from now until the next performance report is due.

Future Time Period
is after the next reporting period.



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14.2f Monitor Communications

Monitor Communications is monitoring and controlling communications throughout the project life cycle to make certain that the information needs of all the project stakeholders are met.

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14.3 Customer Issues

14.3a Manage and Control Quality
14.3b Control Scope

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14.3a Manage and Control Quality

Manage Quality

is the process of using and improving the quality plan and policy to perform tasks that will most likely lead to creating project outputs to customer's satisfaction.

Control Quality

are the activities used to verify that deliverables are of acceptable quality and that they are complete and correct.

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Quality Management and Control Tools

- Flow Chart
- Check Sheet
- Pareto Chart
- Cause-and-Effect Diagram
- Histogram
- Run Chart
- Control Chart

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Exhibit 14.7 (1 of 2)

Project Management and Quality Tools

Tool	Chapter	Description
Charter	3	High-level agreement to start project describing authority, scope, expectations, and resources
Lesson learned	3 and 15	Knowledge from experience captured and shared
Stakeholder analysis	6	Identification and prioritization of stakeholder desires
Communication management plan	6	Document that guides and assigns responsibility for communication with stakeholders
Voice of the customer	7	Captured desired benefits and features in customer's own words
Brainstorming	7	Quick generation of many ideas to identify gaps, issues, roadblocks, or potential solutions
Quality metrics	7	Crisp definition of what and how to measure specific performance
Project risk review	11	Thorough document review to uncover risks
Root cause analysis	11	Technique to discover underlying reason for problem
Cause and effect diagram	11	A visual outline, often resembling a fish skeleton, used to identify and organize possible causes of a stated outcome
Supplier, input, process, output, customer (SIPOC)	12	High-level view of process and stakeholders
Quality audit	12	Structured process to ensure project activities comply with organizational policies

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Exhibit 14.7 (2 of 2)

Tool	Chapter	Description
Benchmarking	12	Identifying and analyzing best practices for improvement ideas
Flow chart	14	A visual model used to show inputs, flow of work, and outputs and to identify possible data collection points for process improvement
Check sheet	14	A simple, structured form used to gather and organize data for analysis
Pareto chart	14	A vertical bar graph used to identify and plot problems or defects in descending order of frequency or cost
Histogram	14	A vertical bar chart used to show the average, extent of varia-tion, and shape of measurements recorded for a process variable
Run chart	14	A special type of scatter diagram in which one variable is time, used to see how the other variable changes over time
Control chart	14	A run chart with process average and control limits used to distinguish between common and special causes of variation

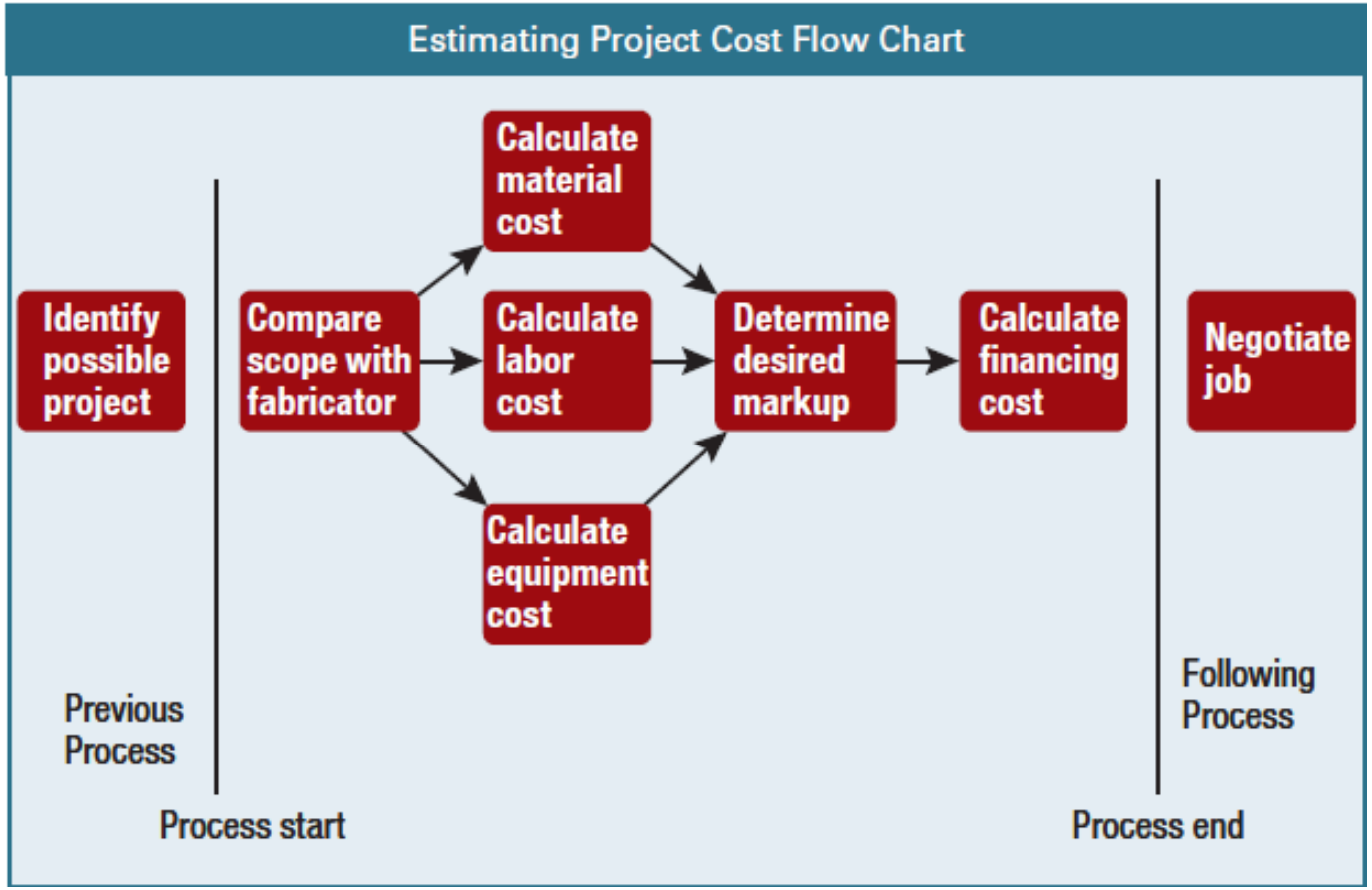
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Flow Chart

- A flow chart is a tool that project managers use as they begin to control quality.
- Flow charts can be used to show any level of detail from the overall flow of an entire project *(such as a network diagram of the project schedule)* down to very specific details of a critical process.
- Flow charts show clearly where a process starts and ends.

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Check Sheet

- Check sheets are customized for each application.
- It is helpful to also collect the date or time when each event happened and notes regarding the impact or any special circumstances.
- When creating categories on a check sheet, it is wise to add a category titled “other” because many times, a problem comes from an unexpected source.

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Exhibit 14.9

Check Sheet for Labor Cost Estimating

Labor Cost Issue	Dollar Impact	Date Discovered	Action Taken
Incorrect scope used			
Category of labor			
Quantity of labor			
Hourly rate			
Pace of labor learning			
Unexpected experience level			
Mathematical error			
Other (be specific)			

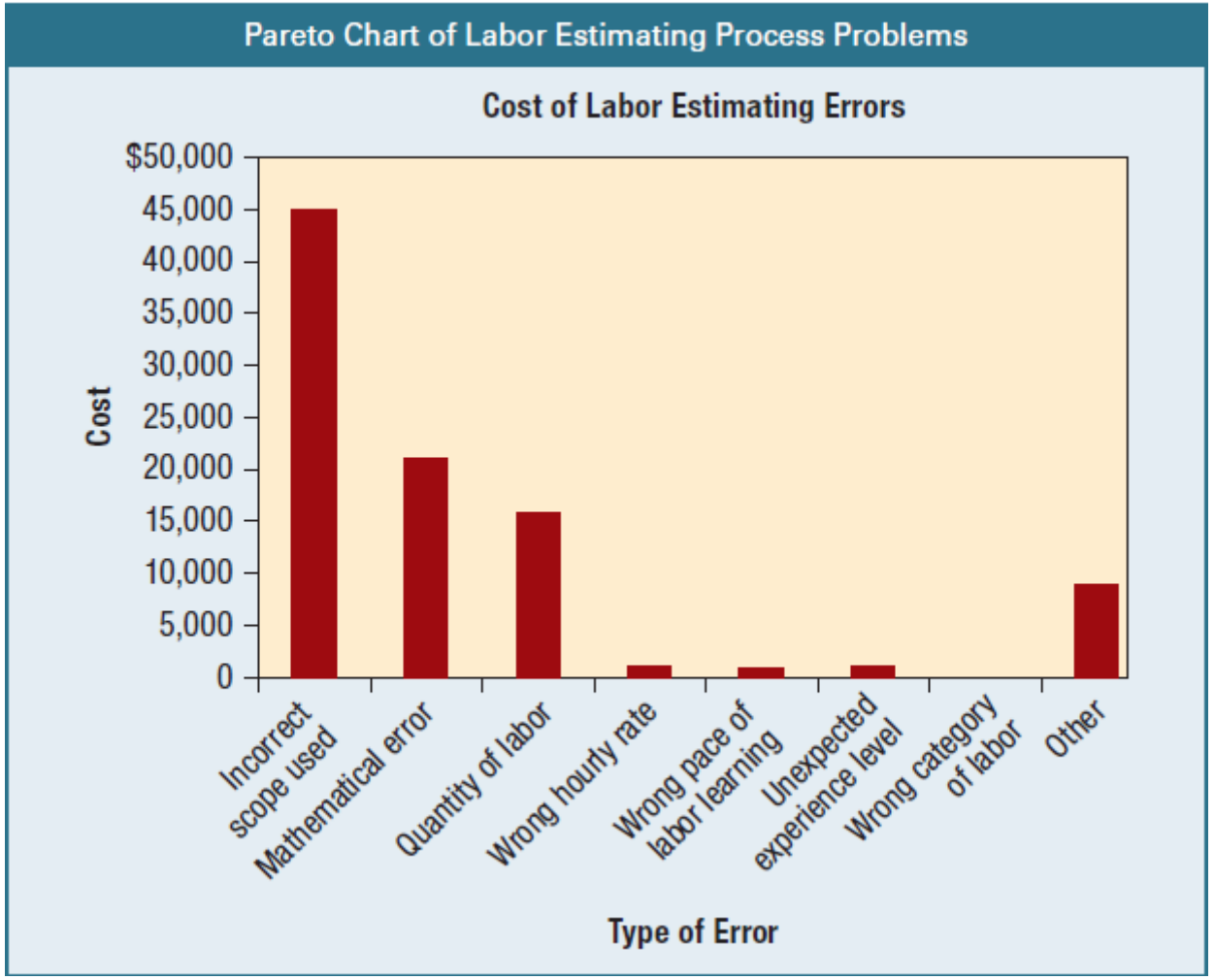
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Pareto Chart

The purpose of the Pareto chart is to quickly understand the primary sources of a problem using the 80/20 rule, wherein 80 percent of defects often come from only about 20 percent of all the sources.

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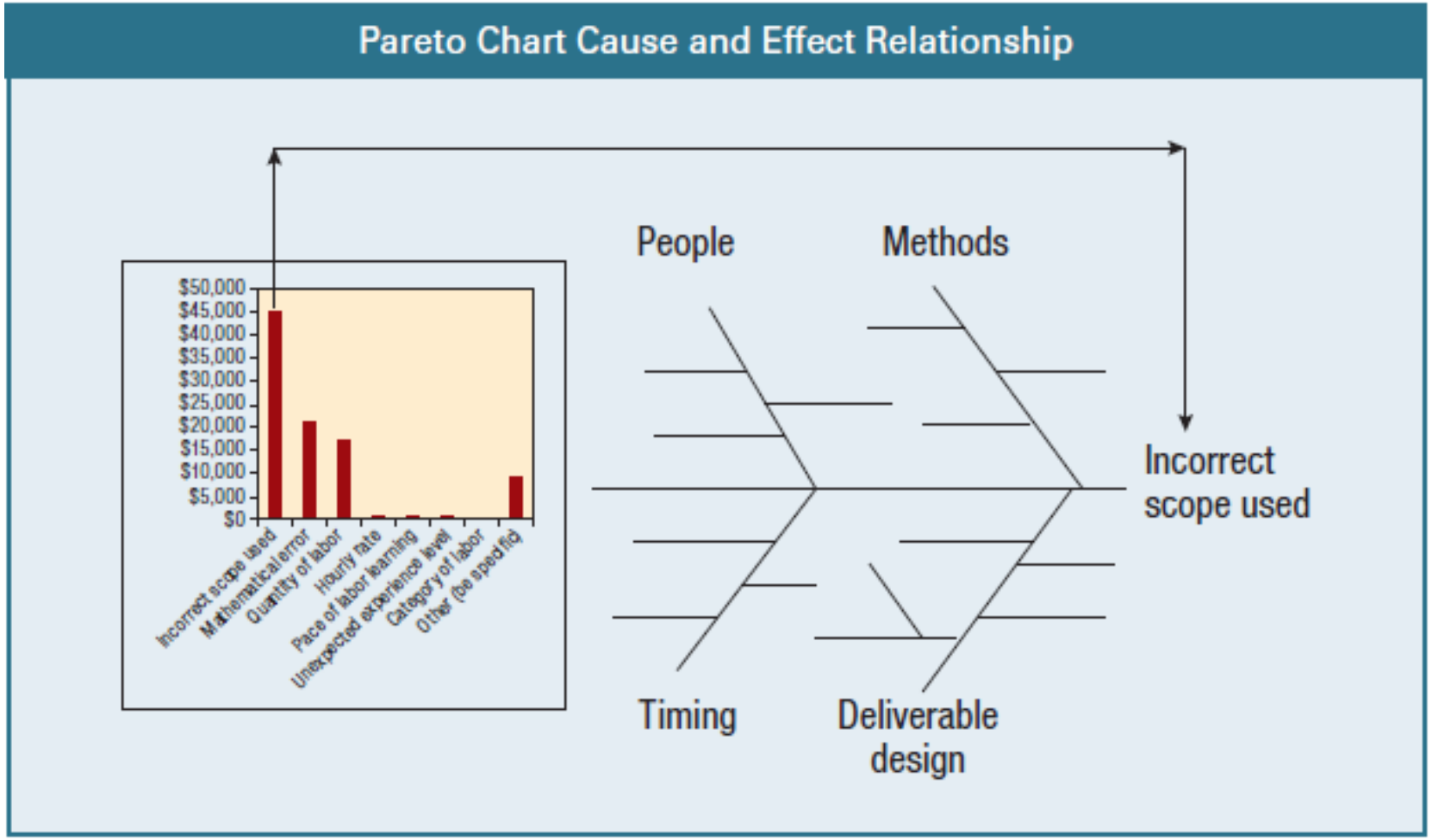
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Cause-and-Effect Diagram

- The **Cause-and-Effect Diagram** *(also known as the Ishikawa diagram)*, is named after its developer, and commonly referred to as the fishbone diagram because it resembles a fish skeleton.
- It is constructed with each “big bone” representing a category of possible causes.

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Exhibit 14.11



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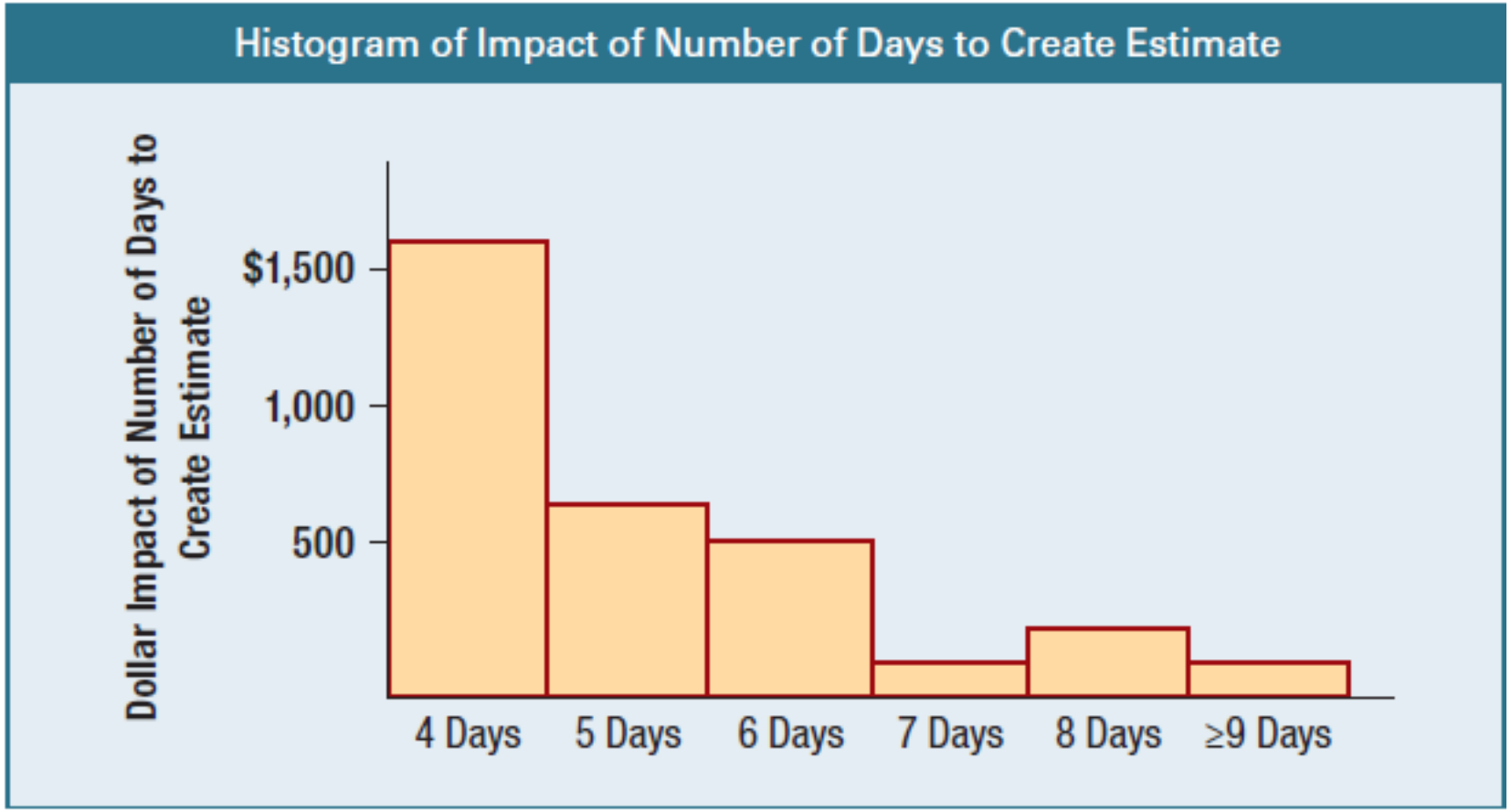
Histogram

A project manager can interpret several things from a histogram such as:

- if nothing unusual is happening, a normal or bell-shaped curve might be expected.
- If the histogram is highly skewed, then there is much more impact happening.

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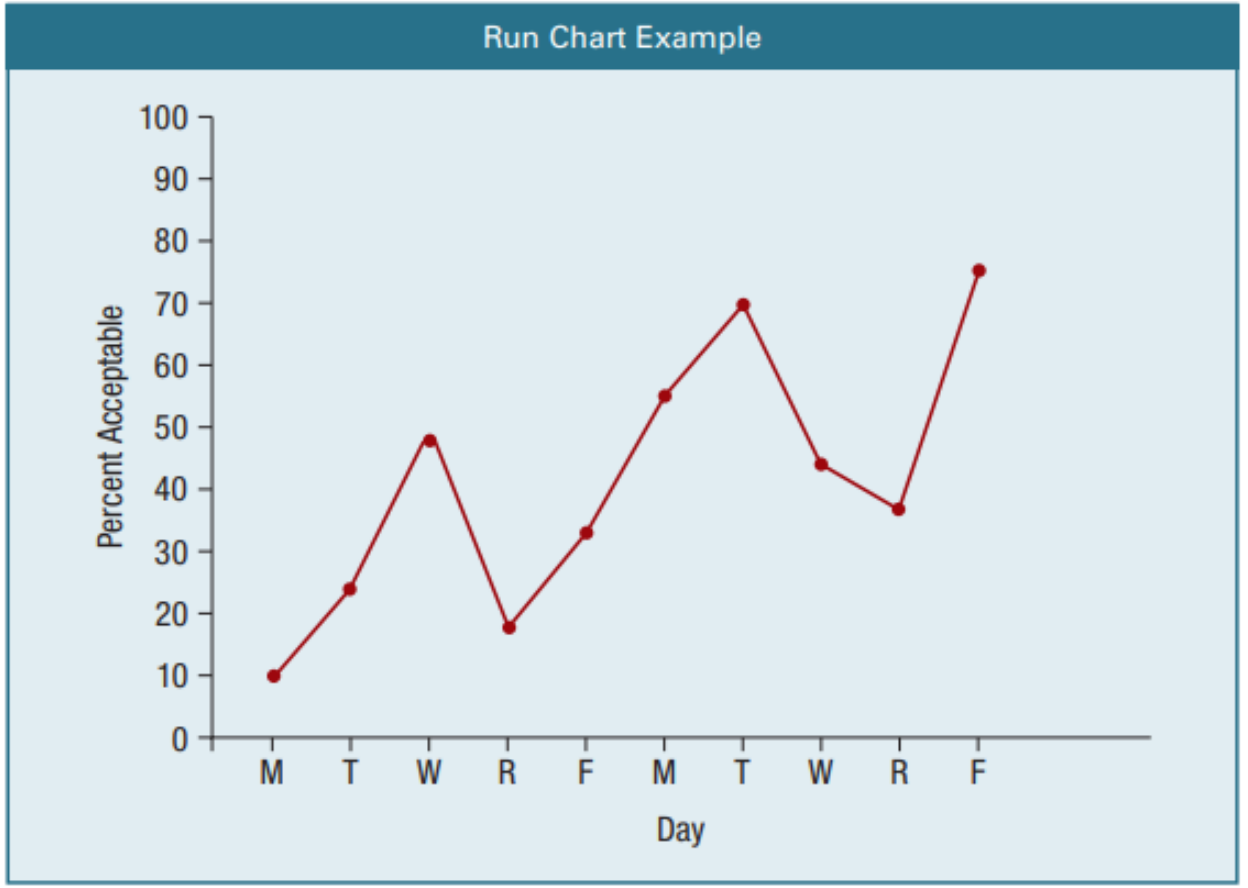
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Run Chart

- Used when the project team wants to see how one specific aspect of the work process may change over time.
- The team could look for three types of variation.
- Is there a trend either up or down?
- Are there abrupt changes, such as either a single point far higher or lower than the others or all of the points suddenly being much higher or lower than previous points.
- How big of a change is big enough to count?

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Exhibit 14.13



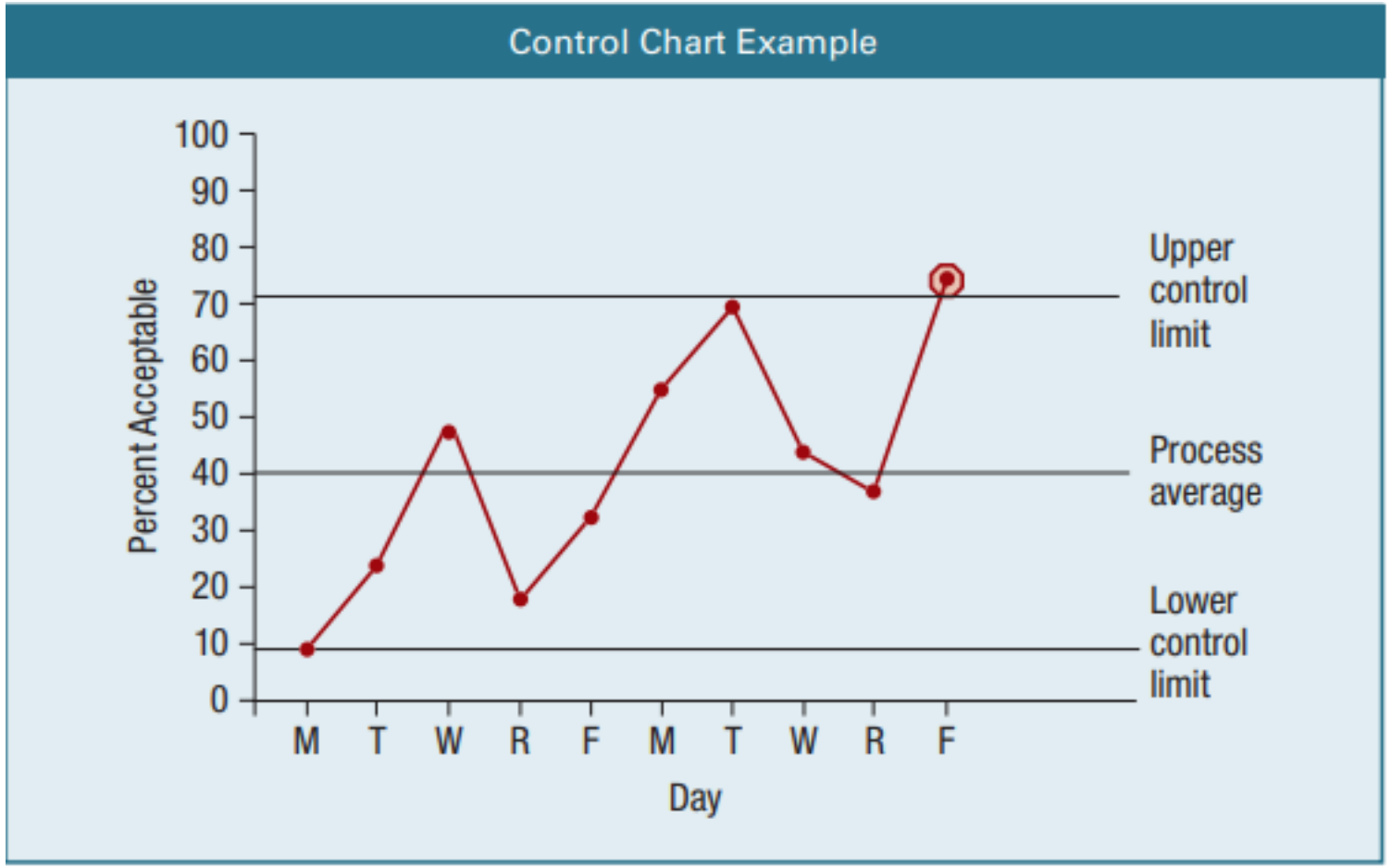
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Control Chart

When considering any of these quality control tools, remember that it is easy to get lost in the details, but the purpose of quality control is to make sure the agreed-upon scope and quality are met per the project charter.

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Exhibit 14.14



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14.3b Control Scope

Control Scope

is the act of closely monitoring the project and product scope status and only allowing necessary changes to the scope baseline.

Scope Change

is any change to the project work activities or deliverable.

Variance Analysis

is the process of determining both the cause and the amount of difference between planned and actual performance.

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14.4 Financial Issues

14.4a Control Resources
14.4b Control Schedule and Costs
14.4c Earned Value Management for Controlling Schedule and Costs

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14.4a Control Resources

Control Resources is a process by which all of the physical resources needed to perform the project are planned and monitored, and changes are made.

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14.4b Control Schedule and Costs

- If the schedule or budget has changed from a previously agreed amount, changes should be formally recommended and managed through the integrated change control system to ensure that impacts on other areas are considered.
- Cost control has one additional consideration—to ensure that no extra money is spent than the authorized amount.

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14.4c Earned Value Management for Controlling Schedule and Costs

Earned Value is used as a decision-making tool.

Earned Value Management is a disciplined and quantitative method to integrate scope, cost, and schedule objectives into a single baseline plan for tracking contract performance.

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Exhibit 14.15

Earned Value Management Terms

Question	Timing	Answer	Acronym
How much work *should be* done?	Now	Planned value	PV
How much work *is* done?	Now	Earned value	EV
How much did *the* "is done" work cost?	Now	Actual cost	AC
How much was the total project *supposed to* cost?	End	Budget at completion	BAC
How much *is* the project schedule ahead or behind?	Now	Schedule variance	SV
How much *is* the project over or under budget?	Now	Cost variance	CV
How efficient is the project *so far* with its schedule?	Now	Schedule performance index	SPI
How efficient is the project *so far* with its budget?	Now	Cost performance index	CPI
How much more do we expect to spend to finish the project?	End	Estimate to complete	ETC
What do we now think the total project will cost?	End	Estimate at completion	EAC
How efficient do we need to be to finish on budget?	End	To-complete performance index	TCPI

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Currently Known Values

Planned value (PV) is “the approved value of the work to be completed in a given time. It is the value that should have been earned as per the schedule.”

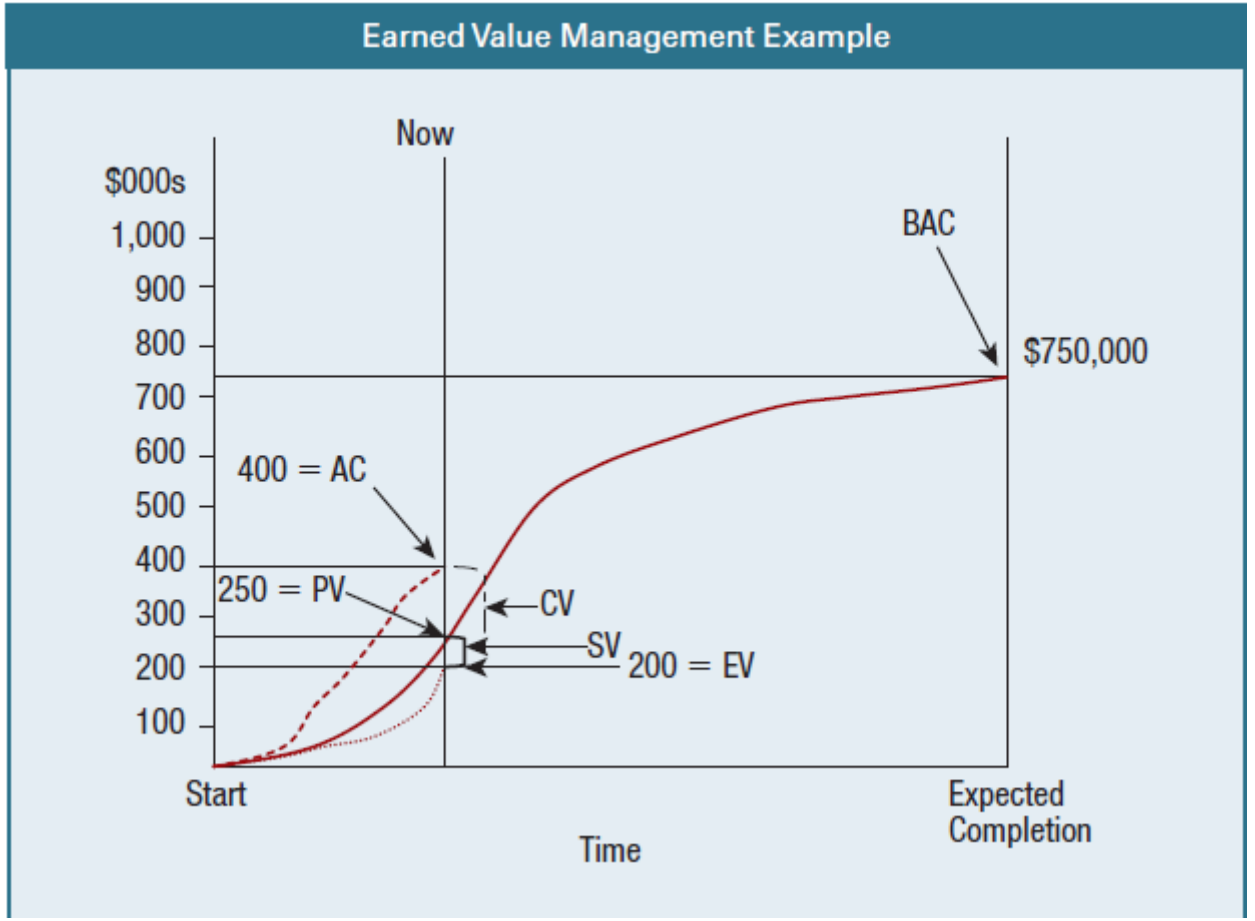
Earned value (EV) is the value of the work actually completed to date.

Actual cost (AC) is the total of costs incurred in accomplishing work on the activity during a given period.

Budget at completion (BAC) is the total amount budgeted for the entire project.

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Exhibit 14.15



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Variances

Schedule Variance (SV) *is the difference between the earned value (EV) and the planned value (PV), and it denotes schedule performance. In our example, it is calculated as $200,000 – $250,000 = – $50,000. We know we are behind schedule because the variation is negative (unfavorable):*

$$SV = EV - PV$$

Cost Variance (CV) *is the difference between earned value (EV) and actual cost (AC), which reflects cost performance. In our example, it is calculated as $200,000 – $400,000 = $200,000. We know we are over budget because the variation is negative (unfavorable):*

$$CV = EV - AC$$

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Indexes

Schedule Performance Index (SPI) *is a schedule performance measure expressed as the ratio of earned value (EV) to planned value (PV). In our example, it is calculated by $200,000/$250,000 = 80%. We know our project is behind schedule because we only accomplished 80 percent of what we planned:*

SPI = EV/PV

With performance indexes, 100 percent means right on plan, less than 100 percent means less efficient than planned, and over 100 percent means more efficient than planned.

Cost Performance Index (CPI) *is a cost performance measure expressed as the ratio of earned value (EV) to actual cost (AC). In our example, it is calculated by $200,000⁄$400,000 = 50%. We know our project is over budget because we have only received $0.50 worth of results for every dollar we have spent:*

CPI = EV/AC

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Estimates (1 of 2)

Estimate to complete (ETC) is the expected budget required to complete all the remaining project work. In our example, if we predict that our future performance will have the same efficiency as our past performance, it is calculated by: (BAC – EV)/CPI = ($750,000 – $200,000)/50% = $1,100,000. First method (Work to date is good estimate of future) ETC = (BAC – EV)/CPI. Unless we improve upon our efficiency, we can expect to pay more for the remaining project work than we originally expected to pay for the entire project! The second method of calculating the ETC is to believe that the original plan is a better predictor than the work to date (maybe because of unusual circumstances that are unlikely to continue). This method is calculated by budget at completion (BAC) – EV = $750,000 – 200,000 = $550,000: Second method (original plan is good estimate of future) ETC = BAC – EV.

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Estimates (2 of 2)

Estimate at completion (EAC) is the total cost of completing all the project work expressed as the sum of actual cost to date and the estimate to complete. In our example, if we believe our efficiency to date is a good predictor of the future, it is calculated by $400,000 + $1,100,000 = $1,500,000. On the other hand, if we believe what happened so far will not be repeated and our original plan is good for the remaining work, it is calculated by $400,000 + $550,000 = $950,000: EAC = AC + ETC Because our cost efficiency is only half of our plan (as we learned from our CPI), unless we become more efficient, we can expect to pay double our original estimate! Even if we match our original plan for the rest of the project, we will still be over the budget in the end. Perhaps our sponsor still wants to know what it would take for us to finish on budget. The to-complete performance index (TCPI) is a measure of the cost performance required to complete the remaining project work within the remaining budget. This is the ratio of the remaining work to the remaining budget and on our example is calculated as ($750,000 – $200,000)/($750,000 – $400,000) = 157%. That means that so far, our cost efficiency as measured by our CPI is 50 percent and we need to suddenly raise it to 157 percent for the remainder of the project to complete on budget! TCPI = (BAC – EV)/(BAC – AC).

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Exhibit 14.16

Project Status Report

Project Health	
X	GREEN: On Track (within cost, schedule, scope parameters)
	YELLOW: In Jeopardy
	RED: Off Track

Weekly Project Status Report

Submitted By	Joe Smith
Role:	Project Manager
Project Name:	Projects and Programs Annual Conference
Report Date:	November 27, 2021

Project Objective
To manage and deliver the annual project and programs conference by March 26, 2022, within a cost of $150,000.

Status	*Include color (green, yellow, red) and explanation*
Overall	
Scope	Green: There are 10 deliverables, including Project Closeout. The team worked on four deliverables this week.
Schedule	Green: The conference is 17 weeks away. We are on track for hitting the conference date.
Budget	Green: $140,000 budget remaining ($10,000 spent on deposit to hold venue facility).

Deliverables	*Tasks completed since last report*
Venue	Signed final contract with hotel.
Speakers	Secured keynote speaker and five track speakers.
Registration	Contacted 10 new volunteers to work the registration table.
Sponsorship	Reviewed design for exhibitor space.

Deliverables	*Tasks working on this week (Progress on these tasks will be stated in next report)*
Venue	Get venue room layout on website and ready for conference brochure.
Speakers	Send contracts to keynote speaker and track speakers.
Registration	Follow-up with new volunteers; secure their time slot for registration table work schedule.
Sponsorship	Reach out to five new sponsors with revised sponsorship package.

Issues/Risks	*State problem, status, and next steps*
Sponsorship	Last year's conference sponsors have been slow to respond. Next Steps: The Sponsorship Committee will review the Sponsorship Package and incorporate additional incentives, plus increase social media activity to generate new sponsors.

Comments	*Include any lessons learned*
Lessons Learned: Do not solely rely on the same sponsors. New sponsors can bring additional products and services of interest to conference attendees. Next week's report will include task activities from Marketing and Staffing Deliverables.	

Source: Connie Plowman, PMP-Retired, co-author of *Project Communications: A Critical Factor for Project Success.*

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14.5 *PMBOK Guide 7e* Determining Project Progress and Results

14.5a Stakeholders
14.5b Team
14.5c Development Approach
14.5d Planning
14.5e Delivery
14.5f Measurement

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14.5a Stakeholders

- The project team works with stakeholders to elicit requirements and expectations about the deliverables.
- Requirements will lead to articulating acceptance criteria, technical performance measures, and definition of done.

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14.5b Team

The project team is responsible for identification, analysis, and engagement of both internal and external stakeholders including those who are supportive of the project and those who are neutral or opposed to the project.

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14.5c Development Approach

Project cadence—*single delivery, multiple deliveries, or periodic deliveries*—dictates the nature and frequency of determining the process of measuring and reporting project progress and results.

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14.5d Planning

Through the project planning process, a team develops an approach to create project deliverables.

Predictive Planning approach starts with high-level deliverables which are defined using a WBS, which is then used to develop cost and schedule estimates.

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14.5e Delivery

Deliverables are released throughout the project or near project completion depending on the development approach used.

Development approach dictates the process and frequency of determining progress and results.

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14.5e Measurement

Measures for progress and results can be in terms of evaluating performance compared to the plan, tracking use of resources, demonstrating accountability, providing information to stakeholders, assessing whether project deliverable are on track or not, focusing on trade-offs, threats, opportunities, and options, and ensuring deliverables meet acceptance criteria.

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14.6 Agile (1 of 2)

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14.6 Comparison of Plan-Driven and Agile Approaches for Determining Project Progress and Results Questions
14.6a Agile Terms

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14.6 Agile (2 of 2)

Comparison of Plan-Driven and Agile Approaches for Determining Project Progress and Results

Determining Project Progress and Results Questions	**Plan-Driven**	**Agile**
How is project work directed and managed?	Work is controlled by using the approved project plan plus changes and corrective actions	Work is prioritized by the PO from the backlog, and the team selects items from the work item pool
How is risk managed?	PM and team track and identify risks and implement resolution strategies	PM and team track and identify risks and implement resolution strategies, plus team manages work flow
How is performance reported within the team?	At fixed intervals or milestones team members report on assigned deliverables	Daily team members report to each other what they did, what they plan, and potential obstacles
How is performance reported beyond the team?	PM reviews, documents, and reports for team on past progress and current and future plans at agreed intervals and makes recommendations	Team demos results at each iteration to PO and other stakeholders, PO reports work via releases or quarterly plan
What tools are frequently used to report progress?	Progress report templates and trend and variance analysis using earned value	Kanban boards and trend and variance analysis using burndown charts
How is quality managed and controlled?	Quality tools	Quality tools plus emphasis on verification and validation
How are potential scope changes handled?	Potential scope changes are formally proposed and evaluated using an integrated change control system to identify impacts	No changes are allowed within an iteration; PO can pivot at end of iteration

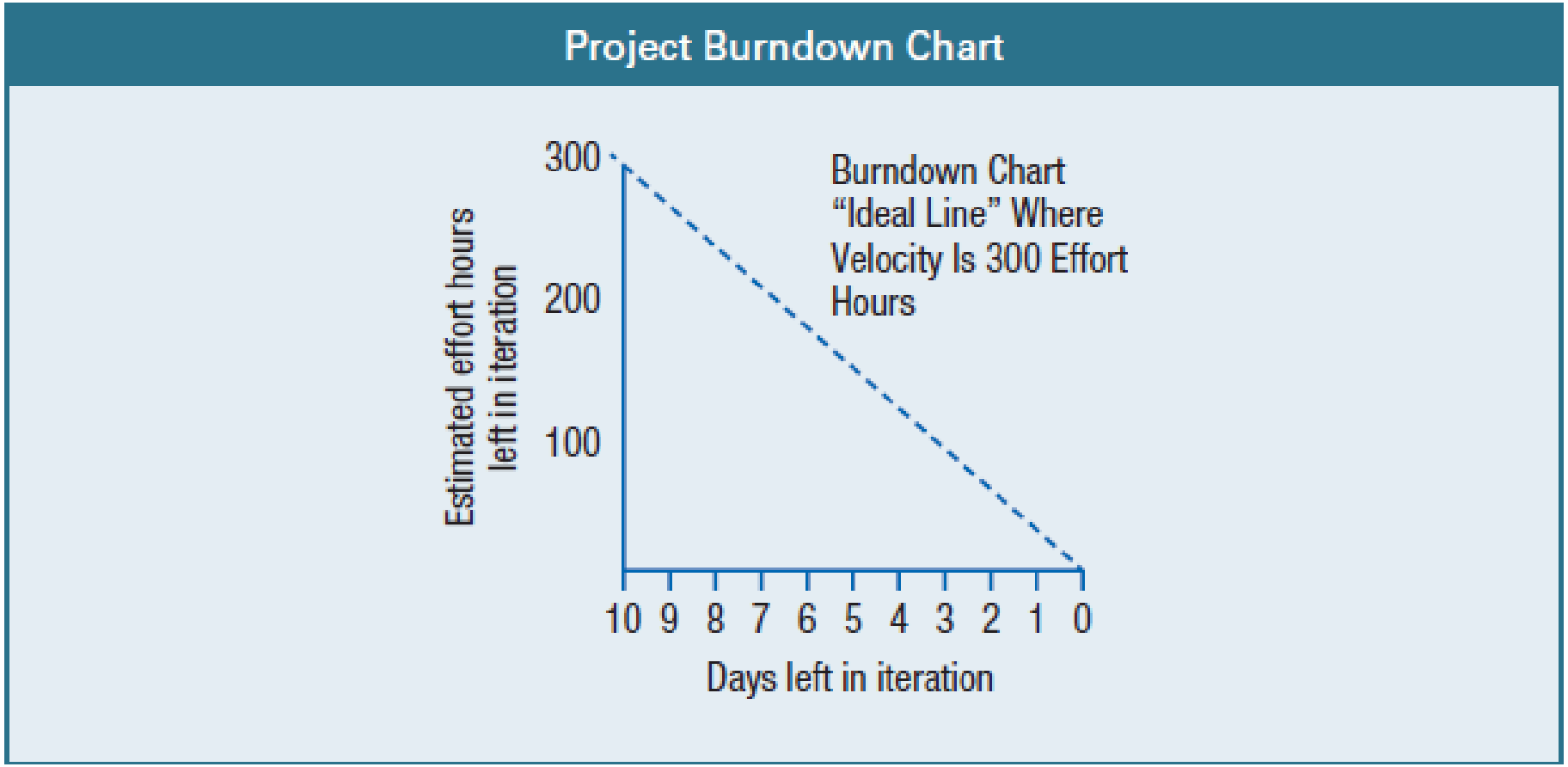
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14.6a Agile Terms

TERM	DEFINITION
Work Item Pool	*All work to do: new requirements, defects, training, and so on.*
Manage Flow	*Teams focus on reducing time to complete work.*
Work-in-Progress (WIP)	*Work that has been started, but has not yet been completed, costs worktime, is not yet of value, and may be obsolete.*
Minimize WIP	*Focus on the work that has been started and finish it promptly.*
Task Switching	*Time and concentration are lost when changing the work a person is doing.*
Waiting	*Delays may mean later feedback and task switching, work is not ready.*
Trend Analysis	*Quantitative review of changes in performance over time.*
Burn-down Chart	*A visual presentation that starts at the total amount of planned work for a sprint and works toward zero.*
Burn-up Chart	*A visual presentation that starts at zero and work their way up toward to the total amount of work expected.*
Pivot	*Changing direction on next sprint based upon customer desires.*

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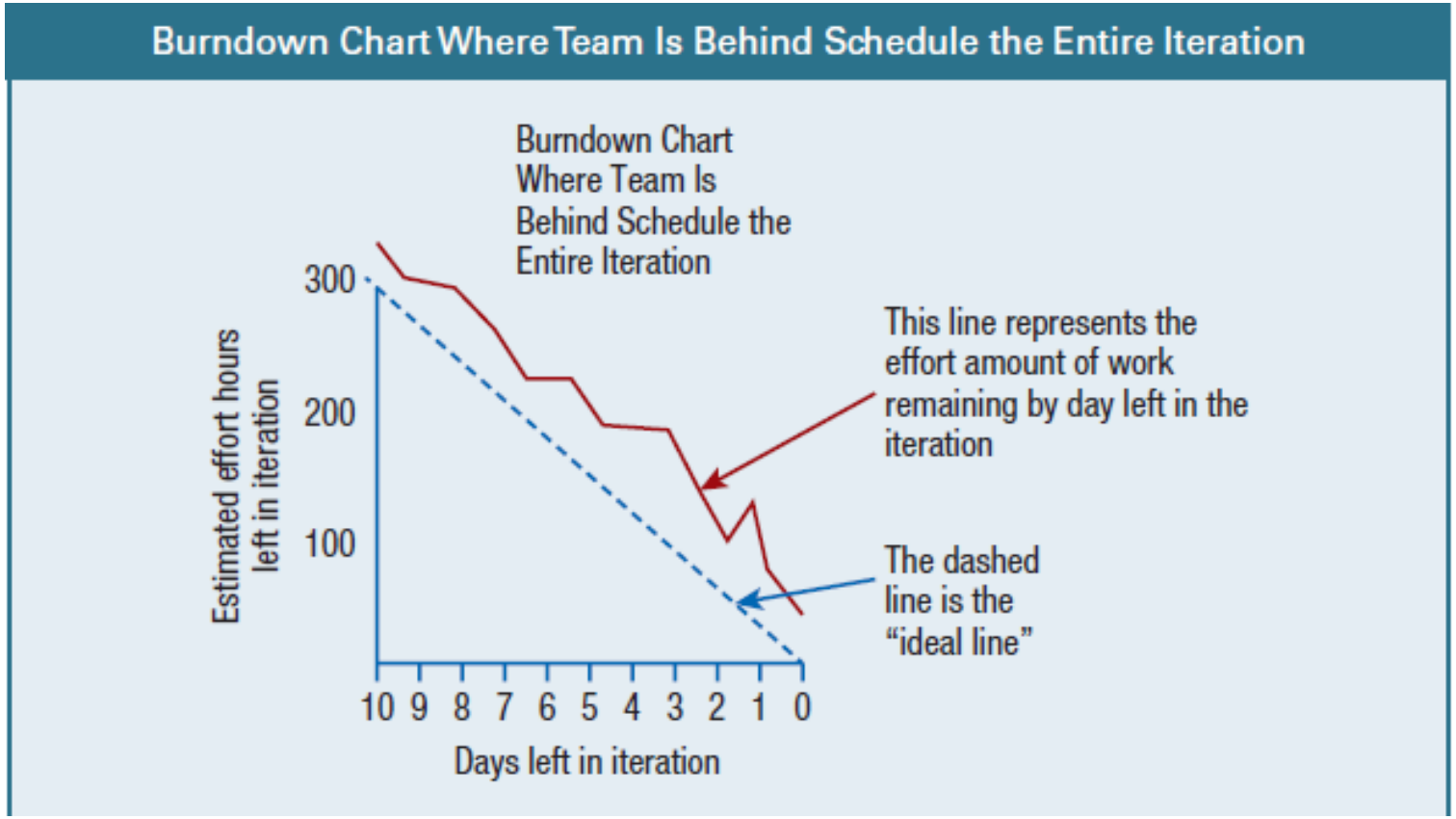
Exhibit 14.17



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Exhibit 14.18

Burndown Chart Where Team Is Behind Schedule the Entire Iteration



Source: Brain Vanderjack, PMP, MBA, CSM, SAFe Certified, author of *The Agile Edge*.

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14.7 Using MS Project to Monitor and Control Projects

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14.7a What Makes a Schedule useful?
14.7b How MS Project Recalculates the Schedule Based on Reported Actuals
14.7c Current and Future Impacts of Time and Cost Variance
14.7d Define the Performance Update Process
14.7e Steps to Update the Project Schedule

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14.7a What Makes a Schedule Useful?

1. **The Baseline Set** *(sometimes called the planned schedule)* a. This set is the original stakeholder-approved scheduled values. b. Data includes the *Baseline Start*, *Baseline Finish*, *Baseline Duration*, *Baseline Work*, and *Baseline Cost*.
2. **The Actual Set** *(sometimes called the performance data)* a. This set is what actually happens during project execution as reported by the resources assigned to project tasks. b. Data includes the *Actual Start*, *Actual Finish*, *Actual Duration*, *Actual Work*, and *Actual Cost*.
3. **The Scheduled Set** a. This set is the future estimated time and costs and is calculated by MS Project. b. Data includes the *Start*, *Finish*, *Duration*, *Work*, and *Cost*. c. Values are continuously recalculated during project execution as tasks and estimates are entered.

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14.7b How MS Project Recalculates the Schedule Based on Reported Actuals

- As actual data is entered into a task's *Actual* field, MS Project copies that data into the task's *Scheduled* field, replacing the estimated values.
- MS Project then recalculates the schedule for future tasks based on a combination of what actually happened and the estimates of the remaining tasks.

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14.7c Current and Future Impacts of Time and Cost Variance

Concepts useful in understanding future impacts of various issues, are:

- Time and cost performance variances from baseline.
- Critical path changes.
- Resource allocation issues.
- Emerging risks.
- Remaining contingency and management reserves.
- The impacts of proposed changes.

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14.7d Define the Performance Update Process

Who Reports? *All team members and suppliers assigned to tasks that were scheduled during the past reporting period need to report. Also, any resource wanting to change the estimate of a soon-to-be-starting task needs to report the new estimate.*

What Is Reported? *Actual Start, Actual Finish, Actual Duration Complete, and Estimated Remaining Duration are reported. The sooner the project manager learns of variances from estimates, the sooner they can take corrective action, making Estimated Remaining Duration and Actual Finish among the most important values to update accurately.*

When to Report? The project manager determines what day of the week resources will report performance *("Status Date" or "As of Date")*, as well as the frequency. The Status Date is usually driven by the date of stakeholder review meetings and the time needed to make adjustments before that meeting.

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14.7e Steps to Update the Project Schedule

1. Acquire the Performance Data
2. Set the Status Date *(as of)*
3. Display the Status Date Line on the Gantt Chart
4. Enter the Duration-Based Performance Data
5. Reschedule Remaining Work
6. Revise Future Estimates

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14.7e Step 1: Acquire the Performance Data

Performance Data is duration-based data. From each resource assignment, collect the date when the task started, how much duration has been completed, how much duration remains, and the actual finish date *(if the task has been finished)*.

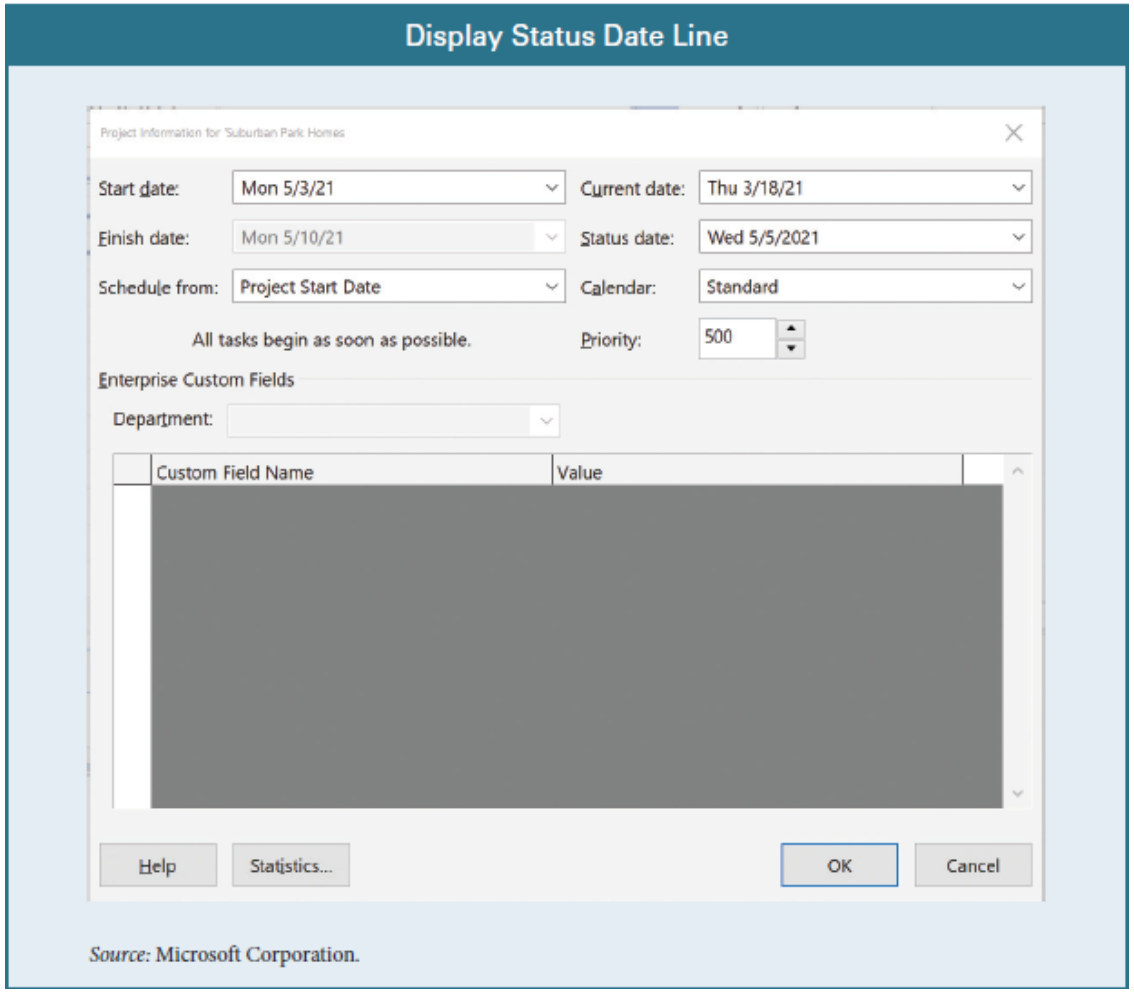
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14.7e Step 2: Set the Status Date (as of)

1. Click the **Project Tab>>Properties Group>>Project Information**.
2. Click the "**Status date"** drop-down.
3. Set status date to 5/5/21 (as shown in Exhibit 14.19).
4. Click **OK**.

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Exhibit 14.19



Source: Microsoft Corporation.

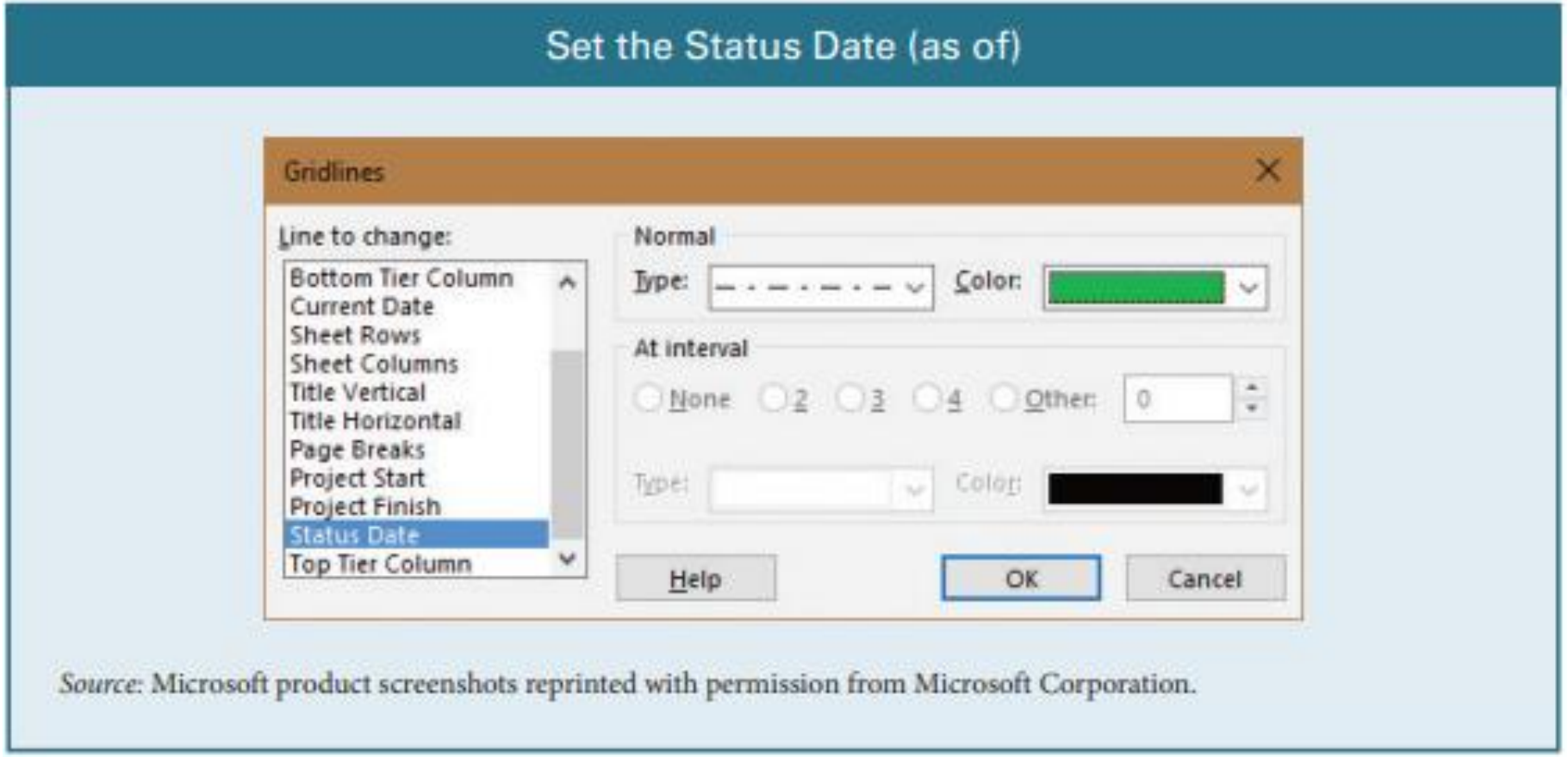
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14.7e Step 3: Display the Status Date Line on the Gantt Chart

1. Click the **Task Tab**>>**View Group**>>**Gantt Chart**.
2. **Right-click** on the right pane>>select **Gridlines**.
3. In the “Line to change list”, select **Status Date**.
4. In the “Normal” box, choose **dashed dotted line**; choose **Green** for the color.
5. Click **OK**.

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Exhibit 14.20



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Step 4: Enter the Duration-Based Performance Data (1 of 5)

SCENARIO 1

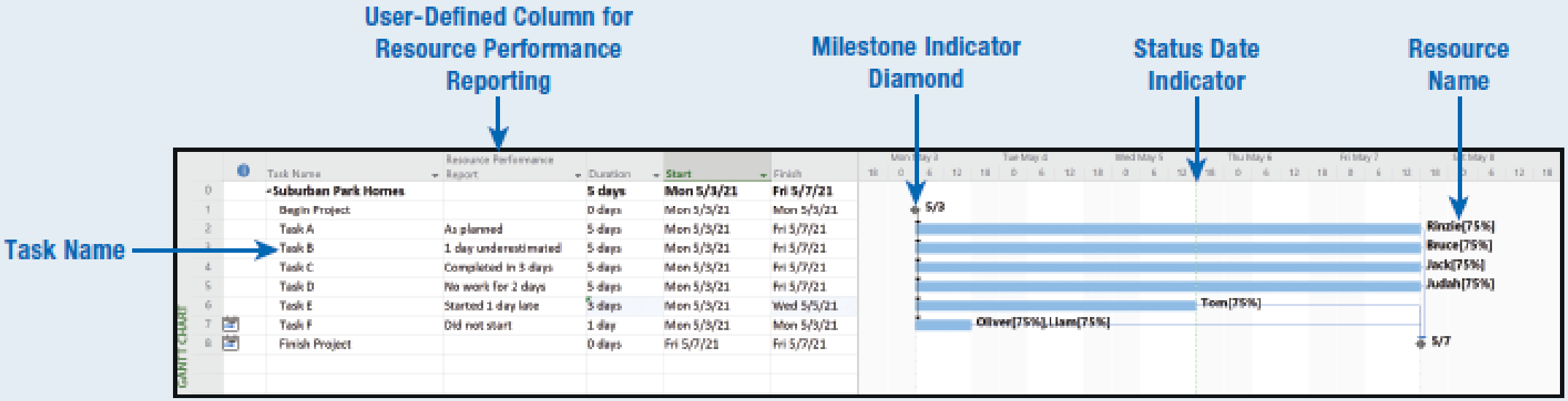
Anniston reports Task A's performance was as scheduled through the end of day on Thursday *(remember that Thursday, May 5 is the Status Date)*.

1. Click **Task A** in the Gantt chart.
2. Click the **Task Tab**>>**Schedule Group**>>click **Mark on Track**.
3. Notice a dark progress bar line appears in the Gantt bar through the end of day on Thursday *(the Status Date)*.

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Exhibit 14.21

Schedule Before Any Actuals Are Applied



Source: Microsoft Corporation.

Cengage

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Step 4: Enter the Duration-Based Performance Data (2 of 5)

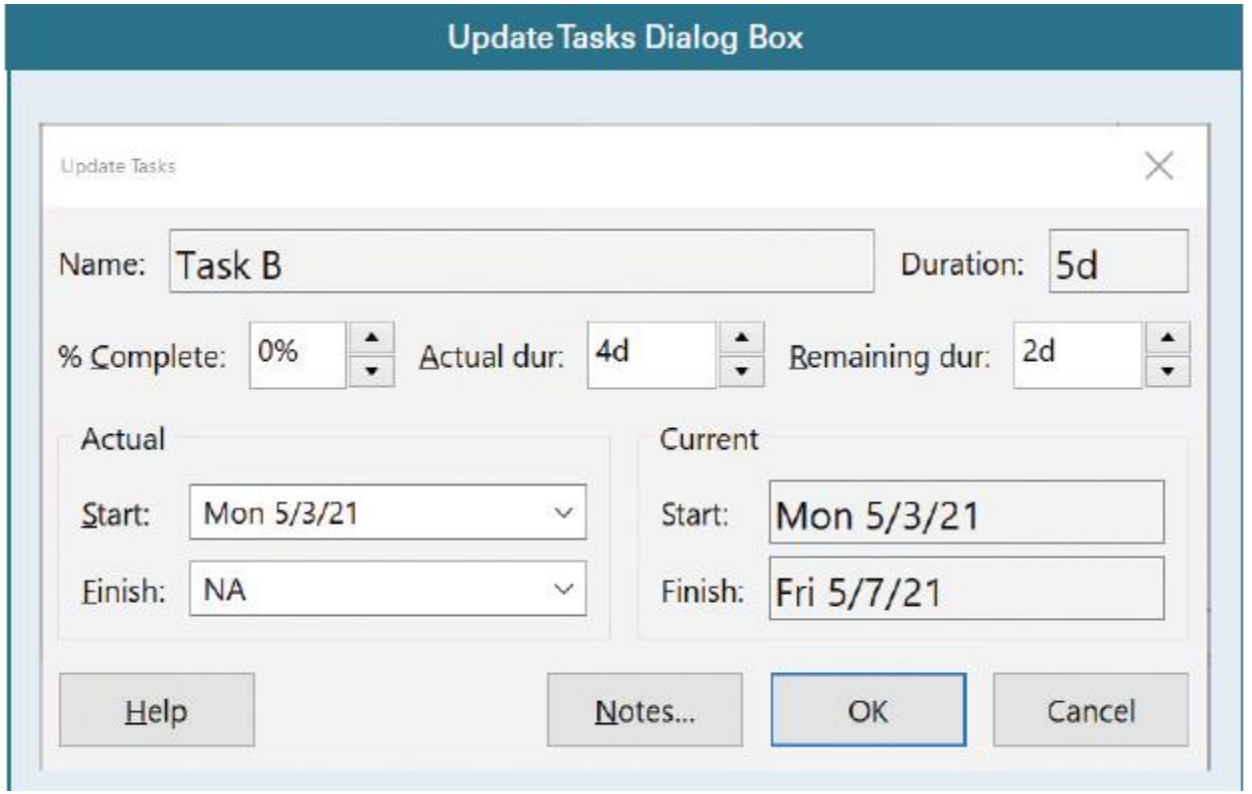
SCENARIO 2

Bruce reports Task B's performance as scheduled, but the estimated remaining duration is two days instead of one.

1. Click **Task B**.
2. Click the **Task Tab**>>**Schedule Group**>>click the **Mark on Track drop-down**>>click **Update Tasks**.
3. In the Update Tasks dialog: a. Actual dur: **enter** "**4d**". b. Remaining dur: **enter** "**2d**" as shown in Exhibit 14.22.
4. Click **OK**.
5. Notice Task B's duration has updated to six days and extends through the end of the day Monday, as shown in Exhibit 14.23.

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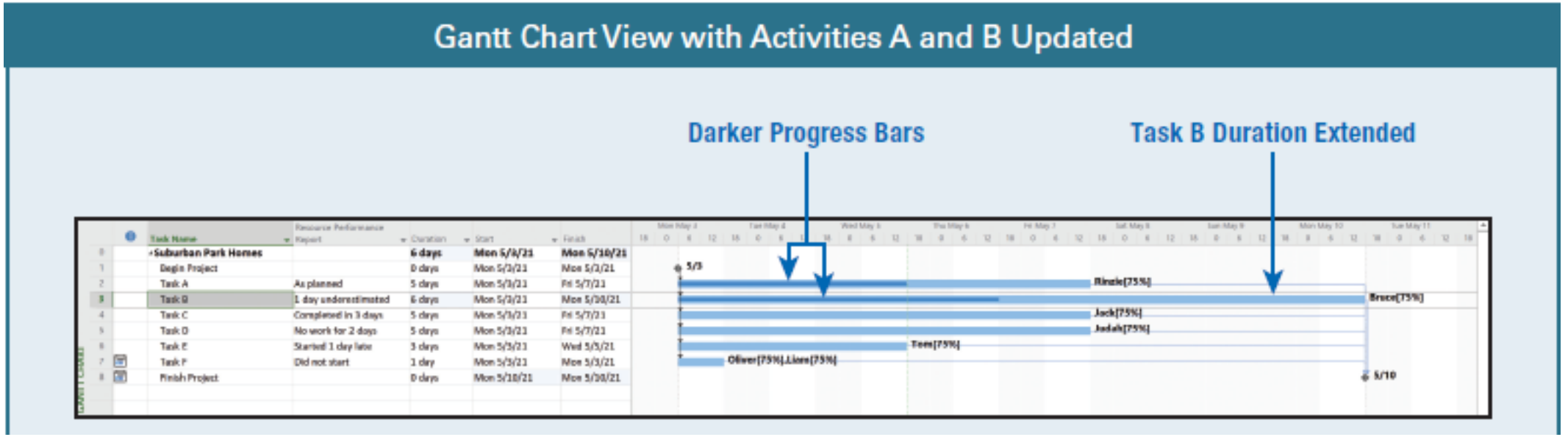
Exhibit 14.22



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Exhibit 14.23



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Step 4: Enter the Duration-Based Performance Data (3 of 5)

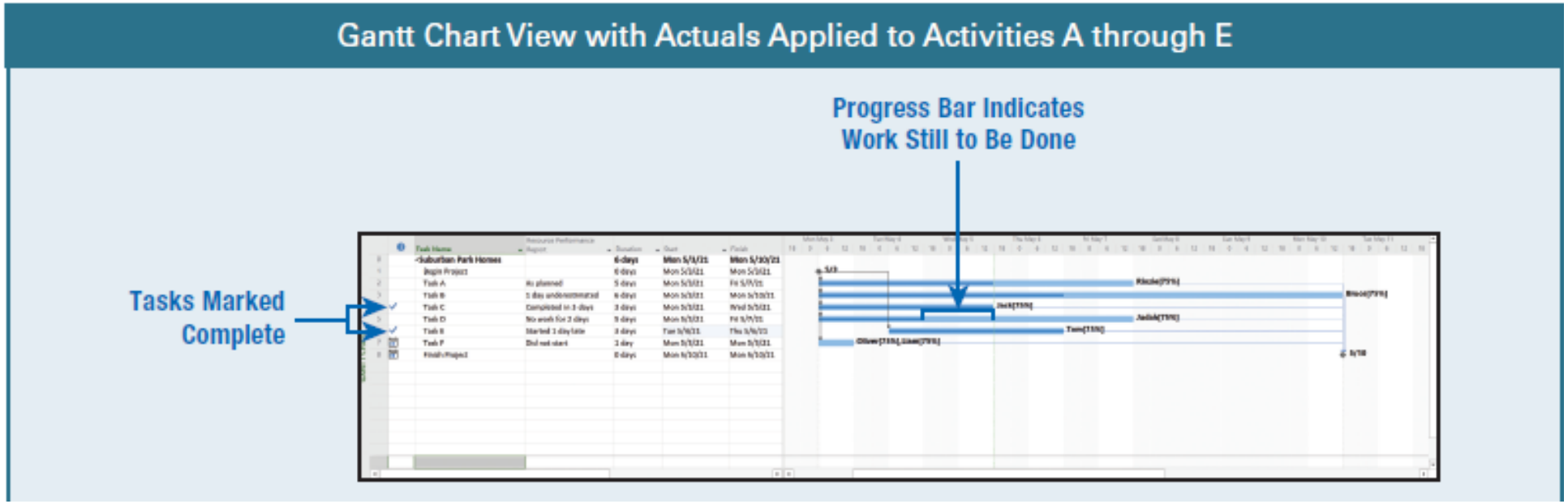
SCENARIO 3

Jack reports that Task C finished two days early.

1. Click **Task C**.
2. Click the **Task Tab**>>**Schedule Group**>>click the **Mark on Track drop-down**>>click **Update Tasks**.
3. In the Update Tasks dialog: a. Actual dur: **enter "3d"**. b. Remaining dur: **enter "0d".**
4. Click **OK**.
5. Notice in Exhibit 14.24 that Task C's duration is now three days and the activity is marked complete *(as denoted by a checkmark in the Indicators column)*.

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Exhibit 14.24



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Step 4: Enter the Duration-Based Performance Data (4 of 5)

SCENARIO 4

Judah reports that no work was done for two of the five days on Task D.

1. Click **Task D.**
2. Click the **Task Tab**>>**Schedule Group**>>click the **Mark on Track drop-down**>>click **Update Tasks.**
3. In the Update Tasks dialog: a. Actual dur: enter **“2d”.** b. Remaining dur: enter **“3d”.**
4. Click **OK**.
5. Notice Task D’s dark progress bar indicates there is still work scheduled for Wednesday and Thursday.

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Step 4: Enter the Duration-Based Performance Data (5 of 5)

SCENARIO 5

Brady reports that Task E started one day late.

1. Click **Task E**.
2. Click the **Task Tab**>>**Schedule Group**>>click the **Mark on Track drop-down**>>click **Update Tasks**.
3. In the Update Tasks dialog: a. Actual Start: **enter " 5/6/21". b.** Actual dur: **enter "3d"**. c. Remaining dur: **enter "0d".**
4. Click **OK**.
5. Notice that Task E is marked complete.
   (*Note: Oliver and Liam report that no work was done on Task F. This update will be addressed in the next step.)*

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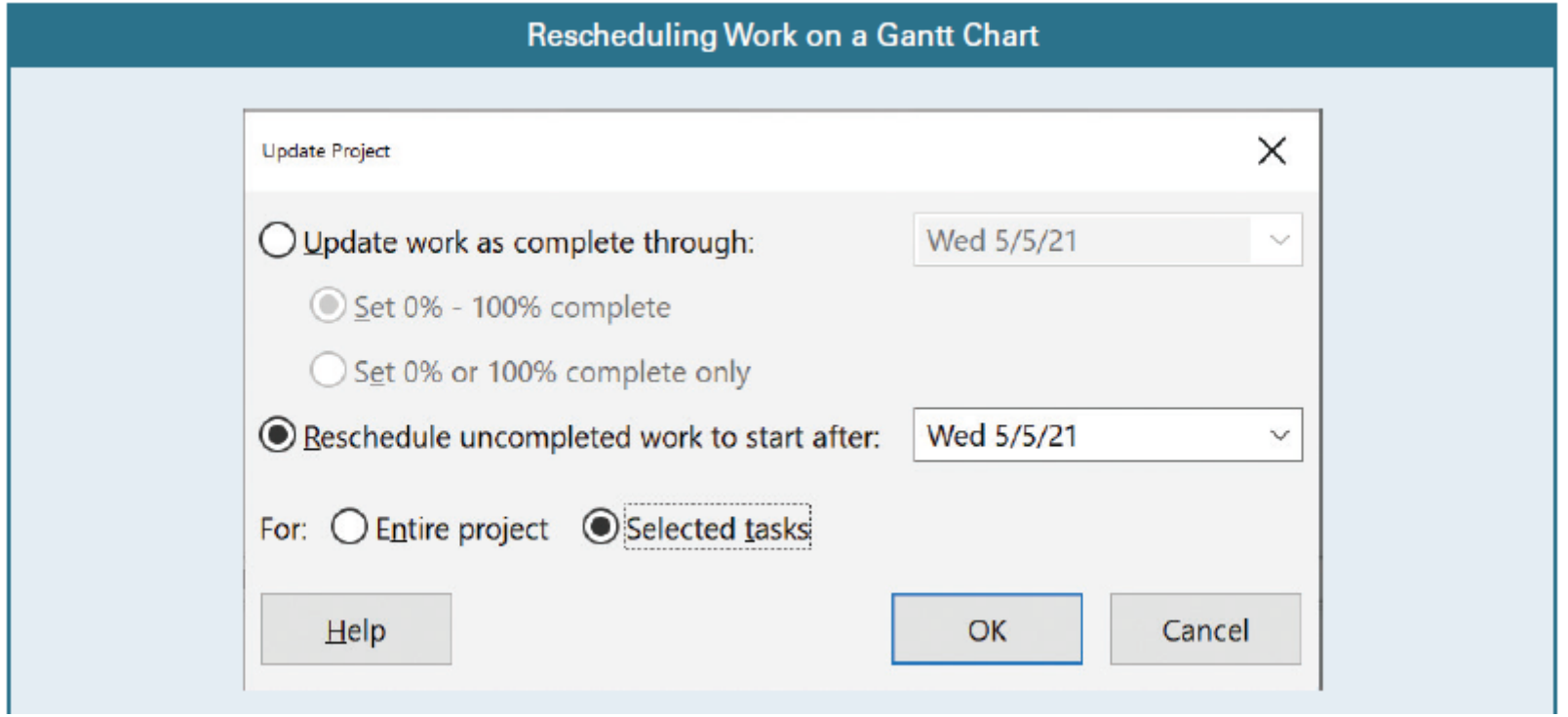
Step 5: Reschedule Remaining Work

SCENARIO 6

1. Click **Task D**.
2. Click the **Project Tab**>>**Status Group**>>click **Update Project.**
3. In the Update Project dialog, click **“Reschedule uncompleted work to start after:”**.
4. Enter the **Status Date** if not already entered as shown in Exhibit 14.25.
5. Click on **Selected tasks**.
6. Click **OK**.
7. Repeat these steps for Task F.

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Exhibit 14.25



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Step 6: Revise Future Estimates

The most accurate estimates are made just before a task gets started. Therefore, at any status meeting, it's a good practice to ask project team members if they believe the estimates for any of their upcoming tasks need updating.

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14.8 Replanning If Necessary

Considerations

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Considerations

Changes a project manager may need to recommend include reassigning activities to different workers, splitting activities so at least some work can get started, reordering activities so they may be accomplished sooner, borrowing or acquiring additional resources, reducing scope.

In Agile projects, replanning is conducted by planning for each release and each iteration. Within an iteration, once the planning is complete, very little additional change is allowed.

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Summary (1 of 2)

- Project manager needs to understand how changes in one area might impact another area.
- Project managers monitor and control project activities.
- Adjustments to projects processed through the project's integrated change control process.
- Potential changes will be proposed, approved/disapproved, documented, and implemented.

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Summary (2 of 2)

- Project managers control project aspects that are subject to tradeoffs—scope, quality, cost, and schedule.
- Use quality control tools to understand quality level, where problems exist, root causes, and how to improve processes.
- Use earned value management (EVM) and MS Project to understand, document, and improve on cost and schedule progress.

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